```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
import pandas as pd
# Load the image
img_path = '/content/F1 🏁🏎 .#f1 #aesthetic #formula1 #racing (00.jpg'
img = Image.open(img_path)

# Convert to numpy array
img_array = np.array(img)

# Perform a numpy operation: Flip the image horizontally
flipped_array = np.flip(img_array, axis=1)

# Display the original and the modified image
fig, axes = plt.subplots(1, 2, figsize=(12, 6))
axes[0].imshow(img_array)
axes[0].set_title('Original')
axes[0].axis('off')

axes[1].imshow(flipped_array)
axes[1].set_title('Flipped (Numpy)')
axes[1].axis('off')

plt.show()
```

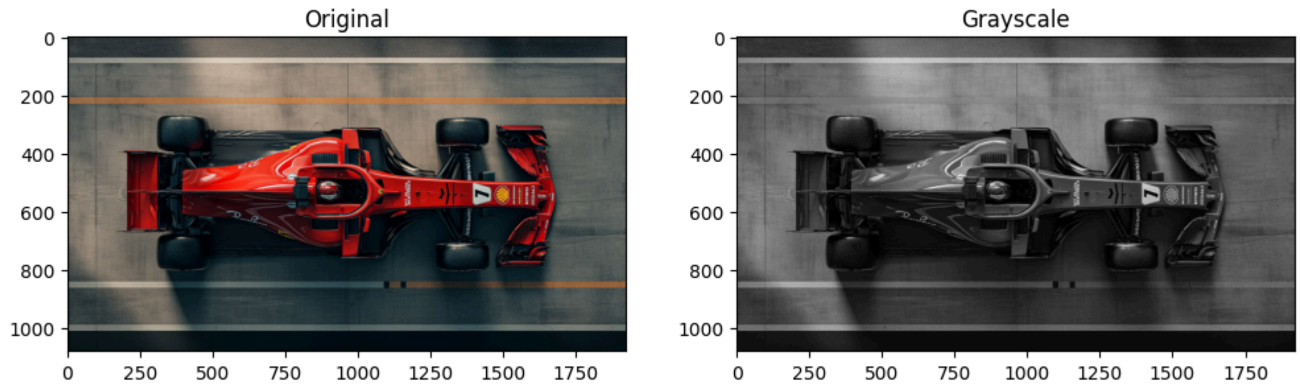
```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipykernel_4533/824692163.py in <cell line: 0>()
      5 # Load the image
      6 img_path = '/content/F1 🏁🏎 .#f1 #aesthetic #formula1 #racing (00.jpg'
----> 7 img = Image.open(img_path)
      8
      9 # Convert to numpy array

/usr/local/lib/python3.12/dist-packages/PIL/Image.py in open(fp, mode, formats)
   3511     if is_path(fp):
   3512         filename = os.fspath(fp)
-> 3513         fp = builtins.open(filename, "rb")
   3514         exclusive_fp = True
   3515     else:

FileNotFoundError: [Errno 2] No such file or directory: '/content/F1 🏁🏎 .#f1 #aesthetic #formula1 #racing (00.jpg'
```

Start coding or generate with AI.

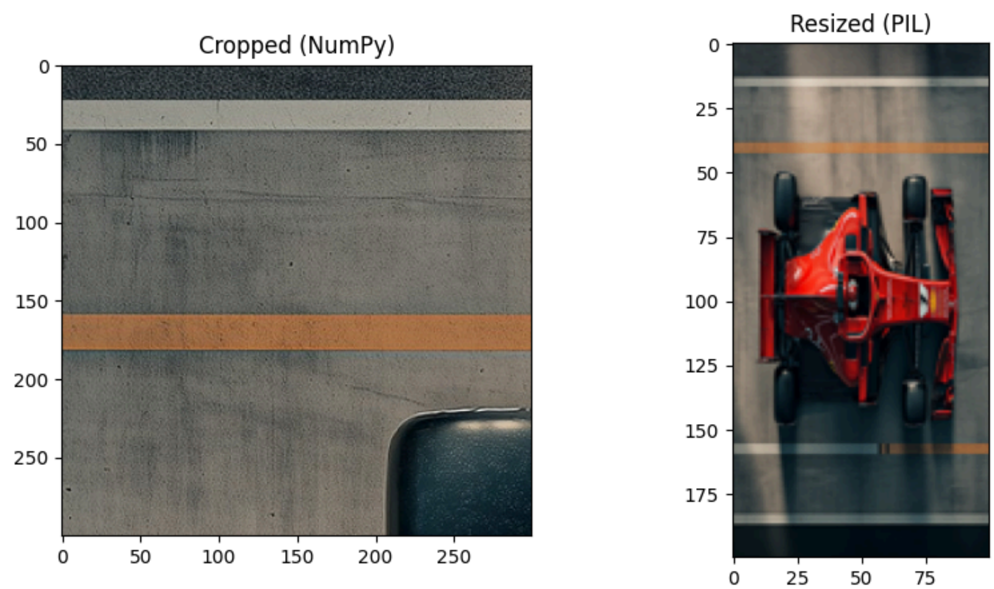
```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt

img_path = '/content/F1 🏁🏎 .#f1 #aesthetic #formula1 #racing (00.jpg'
img = Image.open(img_path)

arr = np.array(img)
print("Shape", arr.shape)
print("Data Type", arr.dtype)
```

```
Shape (1080, 1920, 3)
Data Type uint8
```

```
gray = np.mean(arr, axis=2).astype(np.uint8)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1); plt.title('Original'); plt.imshow(arr)


plt.subplot(1, 2, 2); plt.title('Grayscale');
plt.imshow(gray, cmap='gray')
plt.show()
```

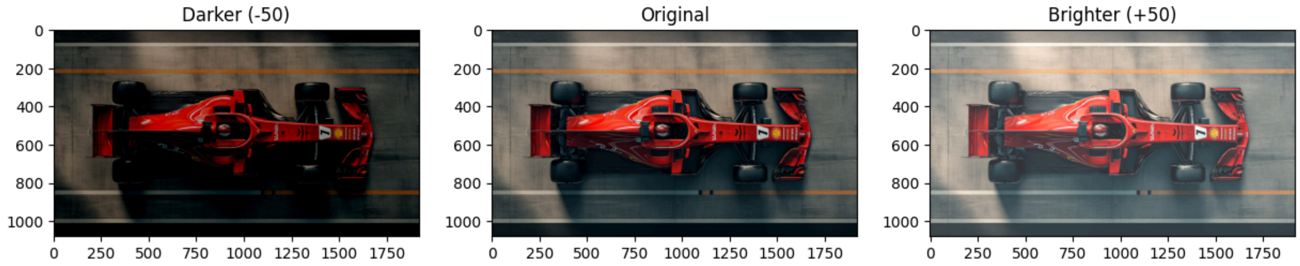



```
# Fix: Ensure the slicing indices are valid (start < end)
crop = arr[50:350, 100:400]
resized = np.array(img.resize((100, 200)))

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.title('Cropped')
plt.imshow(crop)

plt.subplot(1, 2, 2)
plt.title('Resized')
plt.imshow(resized)
plt.show()
```



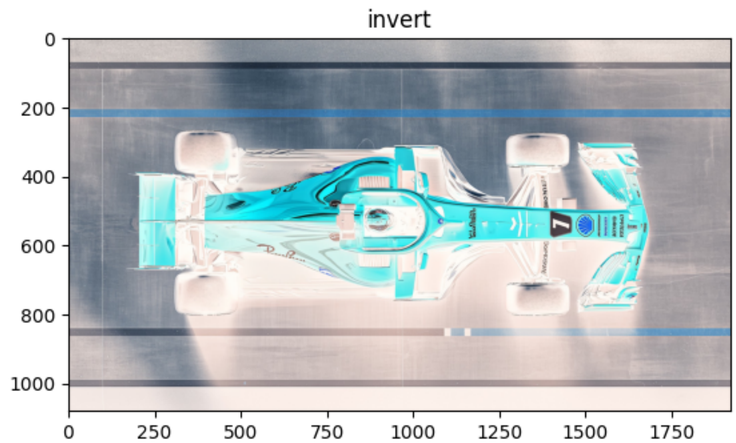
Start coding or generate with AI.

```
brightness_offset = 50

bright_arr = np.clip(arr.astype(np.int16) + brightness_offset, 0, 255).astype(np.uint8)
dark_arr = np.clip(arr.astype(np.int16) - brightness_offset, 0, 255).astype(np.uint8)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1); plt.title('Darker (-50)'); plt.imshow(dark_arr)
plt.subplot(1, 3, 2); plt.title('Original'); plt.imshow(arr)
plt.subplot(1, 3, 3); plt.title('Brighter (+50)'); plt.imshow(bright_arr)

plt.show()
```



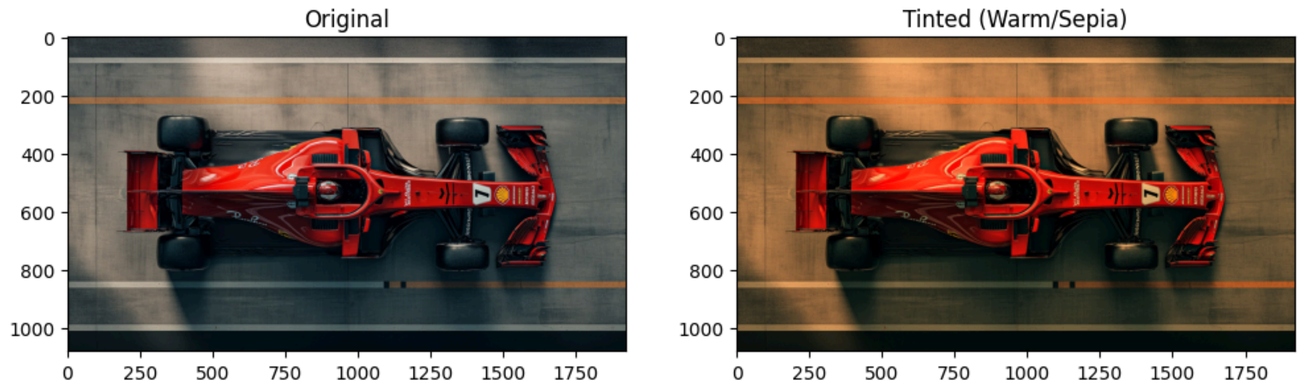

```
invert = 255 - arr
plt.subplot(1, 1, 1); plt.title('invert'); plt.imshow(invert)
```

```
<matplotlib.image.AxesImage at 0x7ed27951f4d0>
```



Start coding or generate with AI.

```
# Define a tint color (e.g., warm/orange tint)
# R, G, B scaling factors
tint_factor = np.array([1.2, 0.9, 0.6])

# Apply the tint by multiplying and then clipping to stay in 0-255 range
tinted_arr = np.clip(arr * tint_factor, 0, 255).astype(np.uint8)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.title('Original')
plt.imshow(arr)

plt.subplot(1, 2, 2)
plt.title('Tinted (Warm/Sepia)')
plt.imshow(tinted_arr)

plt.show()
```




```
import pandas as pd
df1 = pd.DataFrame({"A":[1,2]})
df2 = pd.DataFrame({"A":[3,4]})

result= pd.concat([df1,df2], axis = 1)
print(result)
```

```
   A  A
0  1  3
1  2  4
```

```
G = pd.merge(df1,df2, on = "A",how = "outer")
print(G)
```

```
   A
0  1
1  2
2  3
3  4
```

```
da = pd.DataFrame({"Name": ["Amit", "Jada", "tera", "baap"],
"Subject": ["Math", "Science", "physics", "Chemist"],
"Marks": [78,87,98,65]})
```